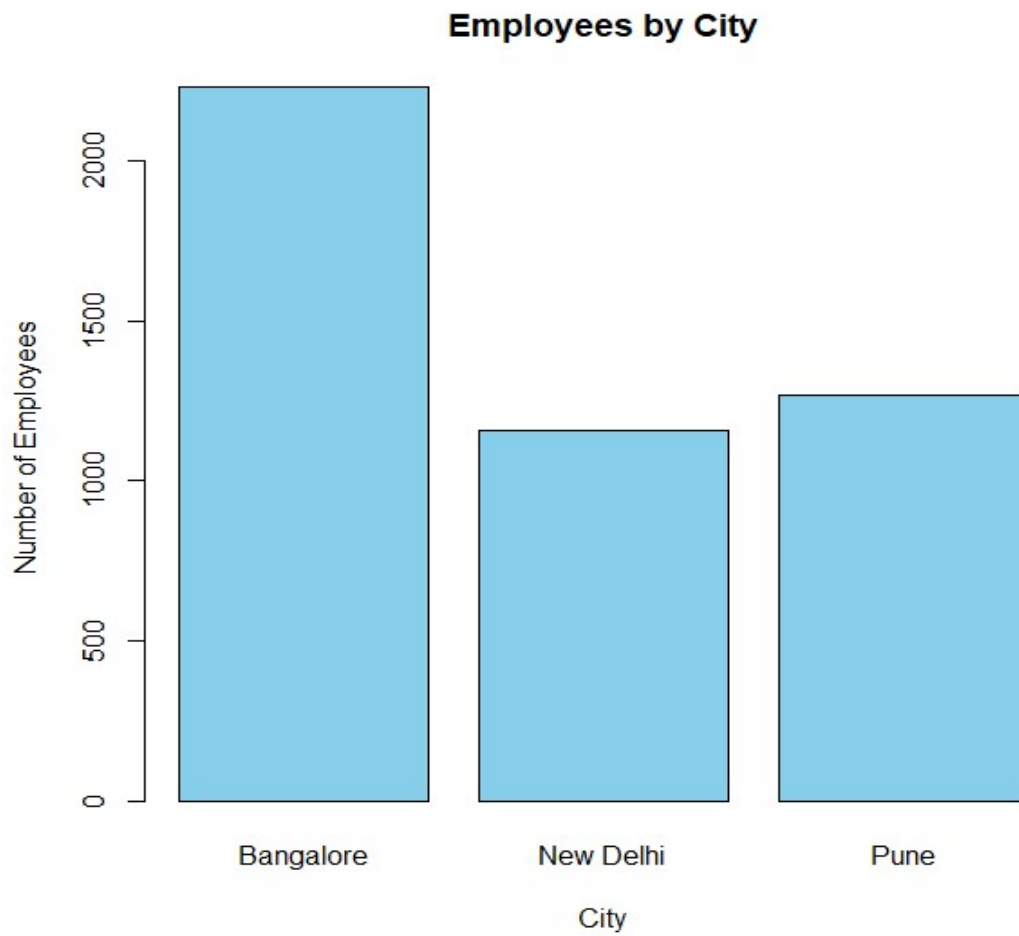


DATA-HANDLING-AND-DATA-VISUVALIZATION**1. Bar Plot****Code**

```
dept <- table(data$City)

barplot(dept,
        col="skyblue",
        main="Employees by City",
        xlab="City",
        ylab="Number of Employees")
```

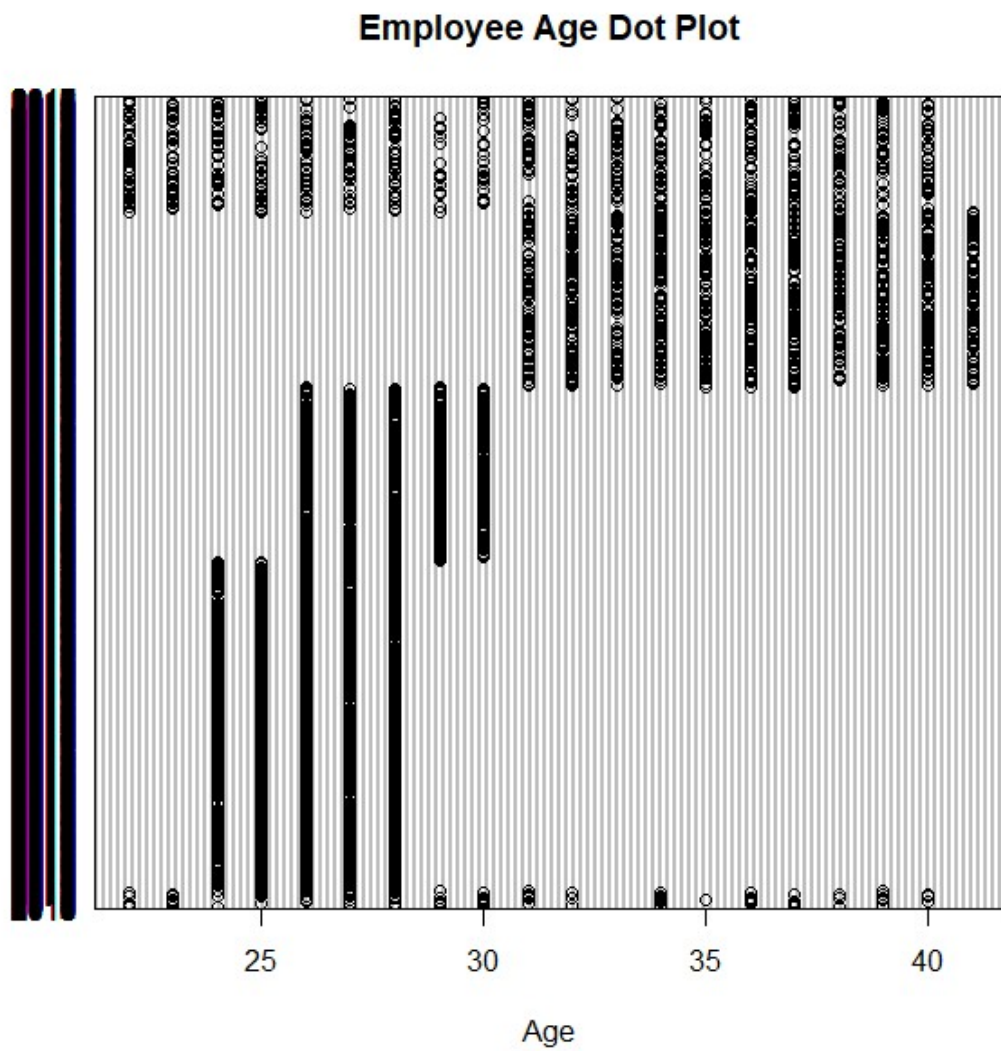
OUTPUT:

2. Dot Plot

Code

```
dotchart(data$Age,  
         labels=data$JoiningYear,  
         main="Employee Age Dot Plot",  
         xlab="Age")
```

OUTPUT:

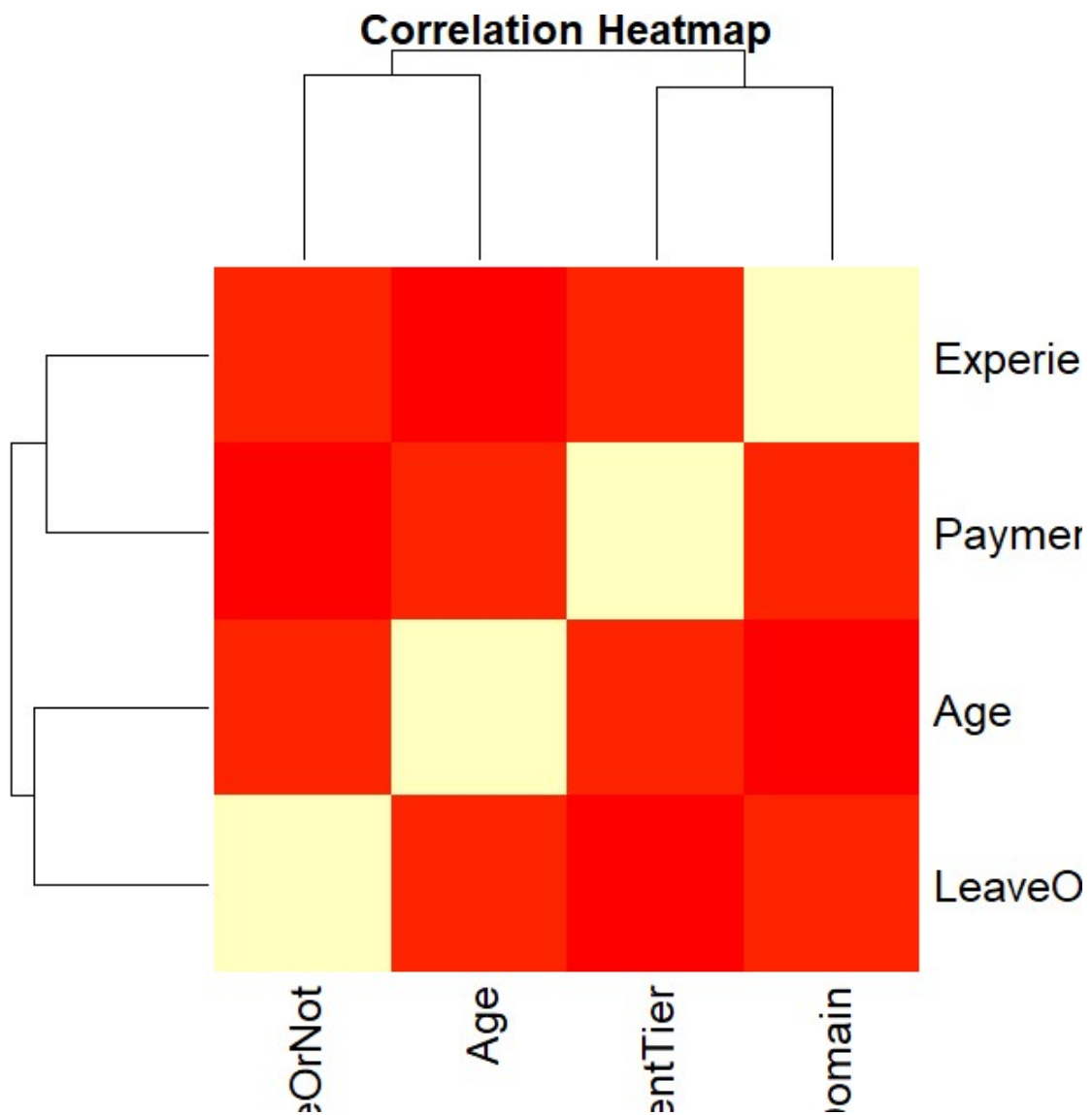


3. Heatmap

Code

```
numeric_data <- data[,c("Age","PaymentTier","ExperienceInCurrentDomain","LeaveOrNot")]  
heatmap(cor(numeric_data),  
        col=heat.colors(10),  
        main="Correlation Heatmap")
```

OUTPUT:

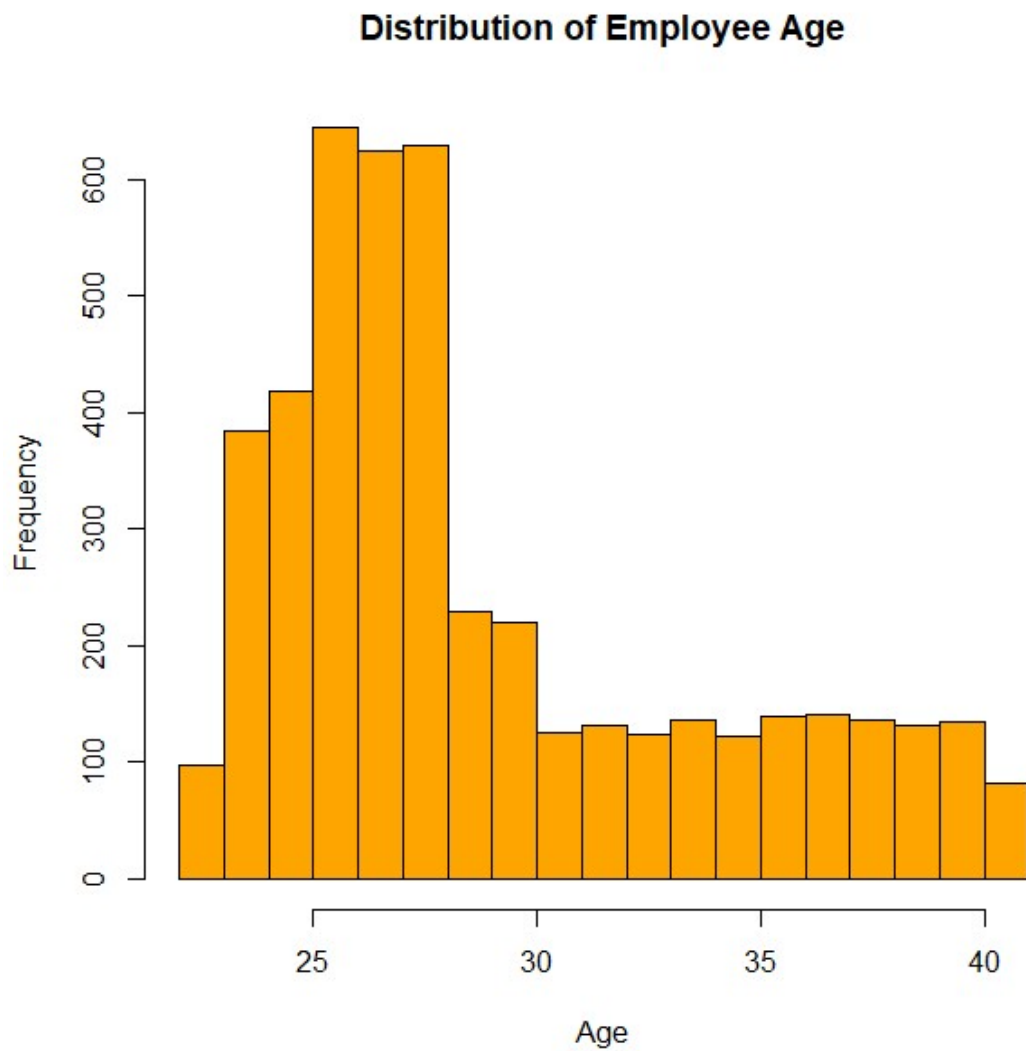


4. Histogram

Code

```
hist(data$Age,
      col="orange",
      main="Distribution of Employee Age",
      xlab="Age",
      ylab="Frequency")
```

OUTPUT:



5. Violin Plot

```
install.packages("vioplot")
```

Code

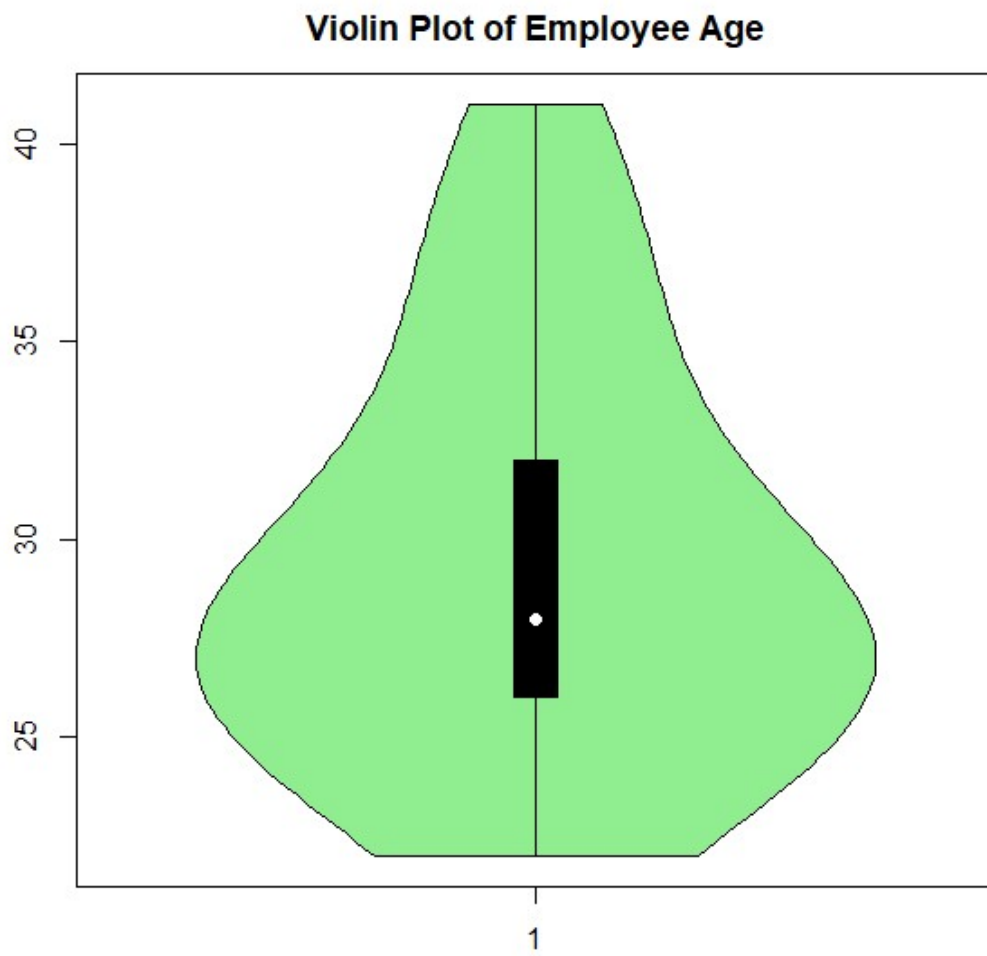
```
library(vioplot)
```

```
vioplot(data$Age,
```

```
       col="lightgreen",
```

```
       main="Violin Plot of Employee Age")
```

OUTPUT:



6. Ridgeline Plot

```
install.packages("ggridges")
```

```
install.packages("ggplot2")
```

Code

```
library(ggplot2)
```

```
library(ggridges)
```

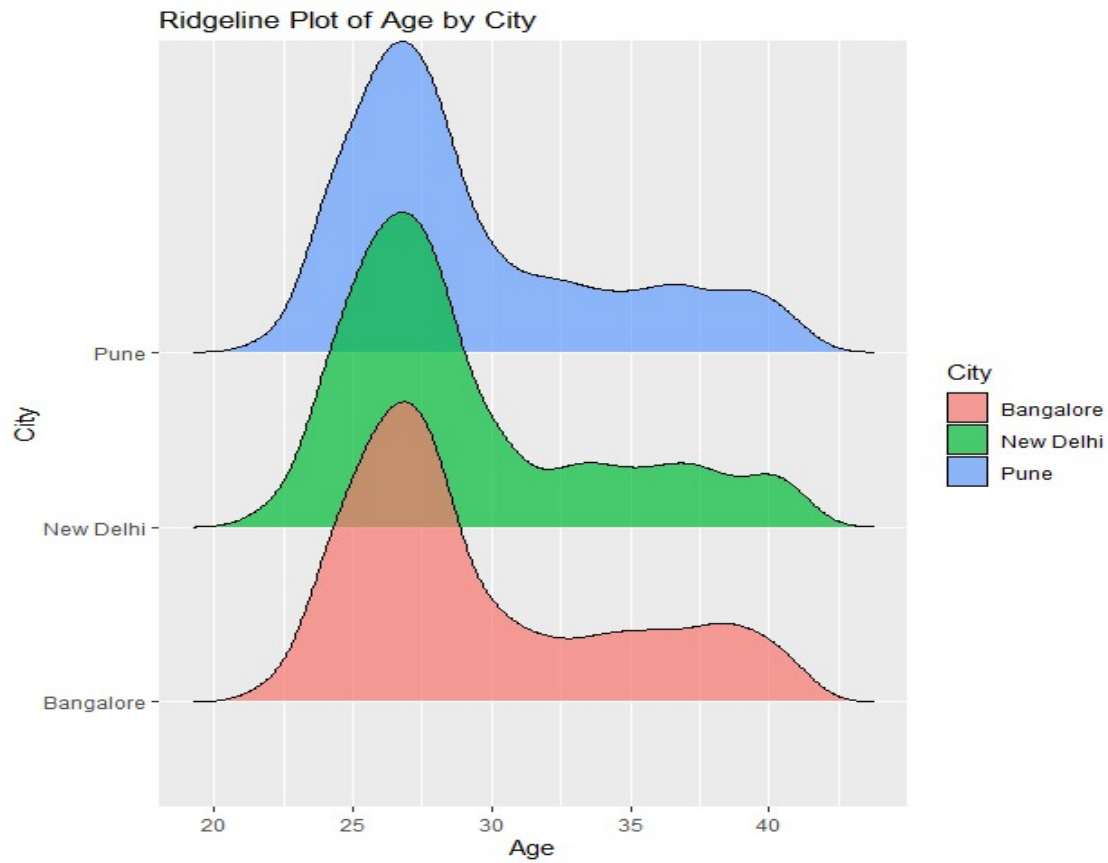
```
ggplot(data,
```

```

aes(x=Age,
    y=City,
    fill=City))+
geom_density_ridges(alpha=0.7)+
labs(title="Ridgeline Plot of Age by City")

```

OUTPUT:



7. Box Plot

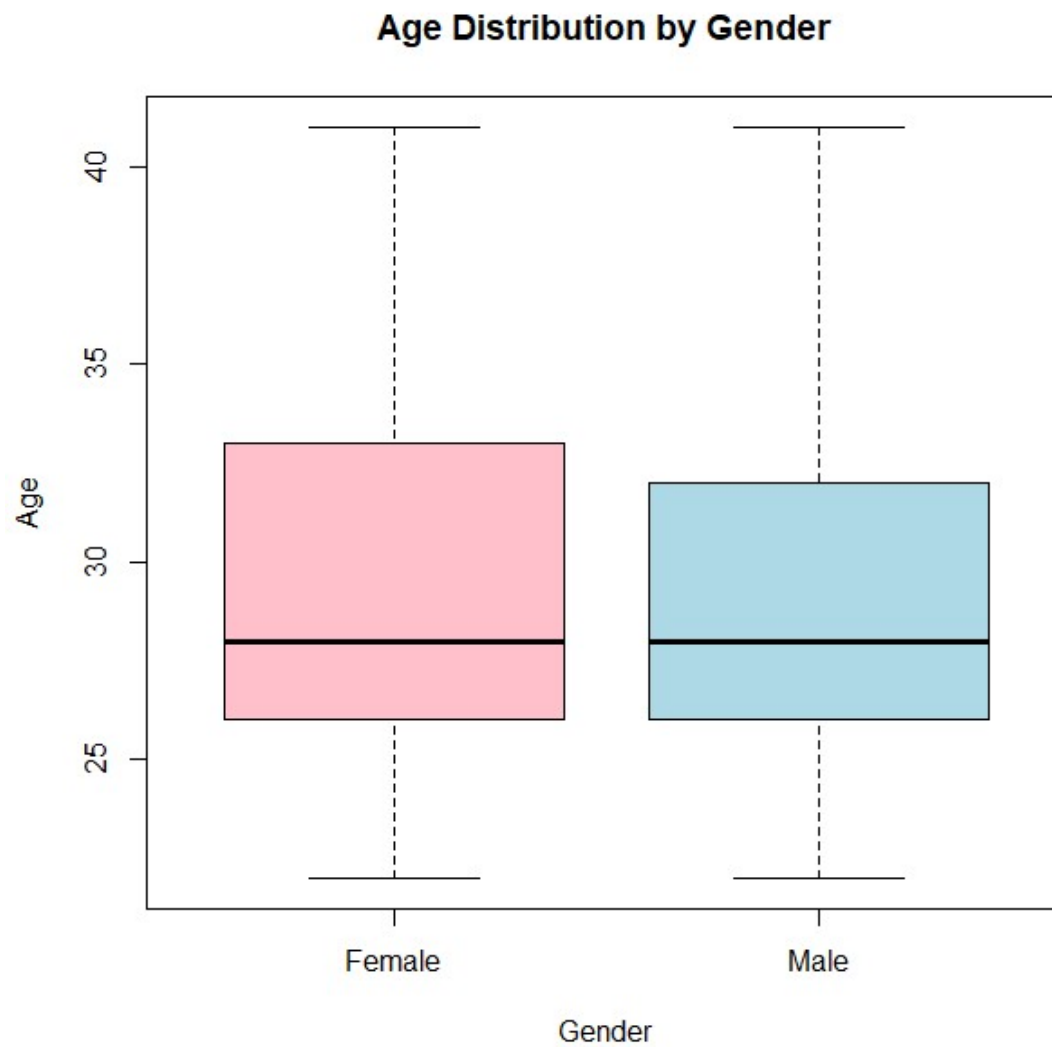
Code

```

boxplot(Age~Gender,
    data=data,
    col=c("pink","lightblue"),
    main="Age Distribution by Gender",
    xlab="Gender",
    ylab="Age")

```

OUTPUT:

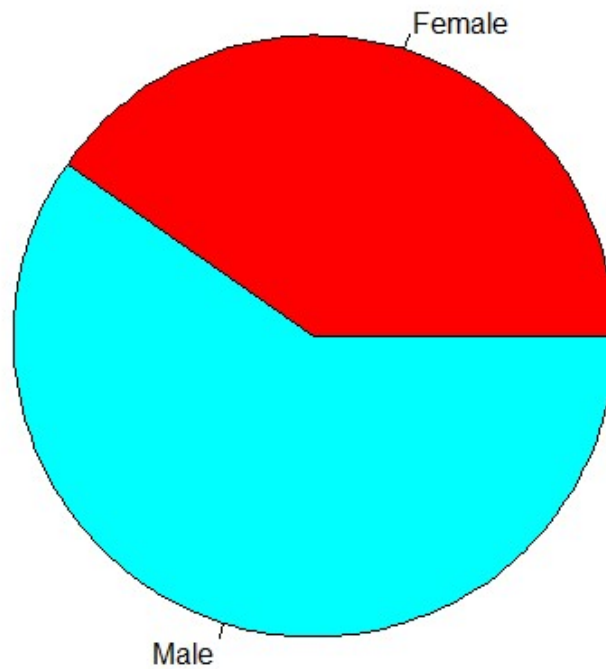


8. Pie Chart

Code

```
gender<-table(data$Gender)
pie(gender,
    labels=names(gender),
    col=rainbow(length(gender)),
    main="Gender Distribution")
```

Gender Distribution



9. Stacked Bar Chart

Code

```
stack<-table(data$City,data$Gender)
barplot(stack,
  beside=FALSE,
  col=c("pink","lightblue"),
  legend=rownames(stack),
  main="Employees by City and Gender",
```

```
xlab="City",  
ylab="Count")
```

OUTPUT:



10. Treemap

```
install.packages("treemap")
```

Code

```
library(treemap)
```

```
treemap(data,
```

```
  index="City",
```

```
  vSize="Age",
```

```
  title="Treemap of Employee Age by City")
```

OUTPUT:

